

CSE 4303

Data Structures

Topic: Graph Basics

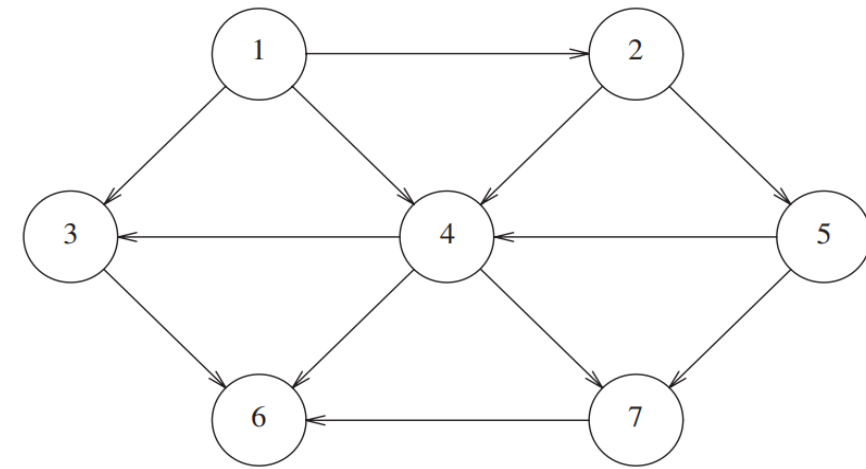
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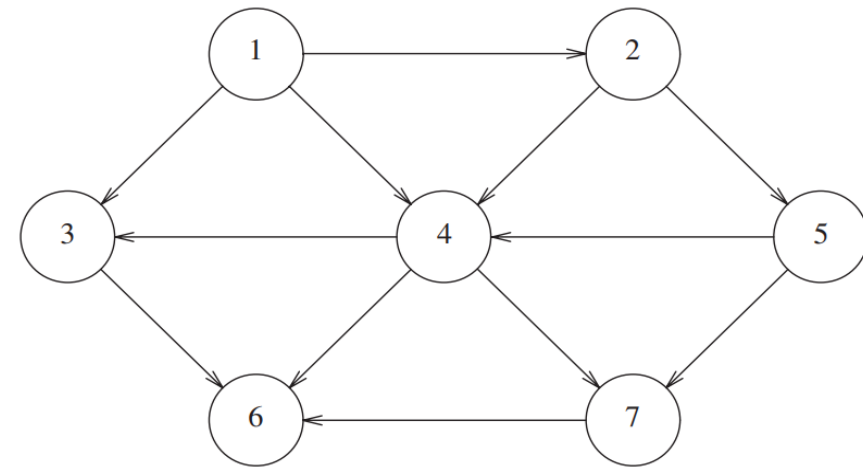
Introduction

- A **Graph** $G = (V, E)$ consists of
 - A set of **Vertices** (V)
 - A set of **Edges** (E)



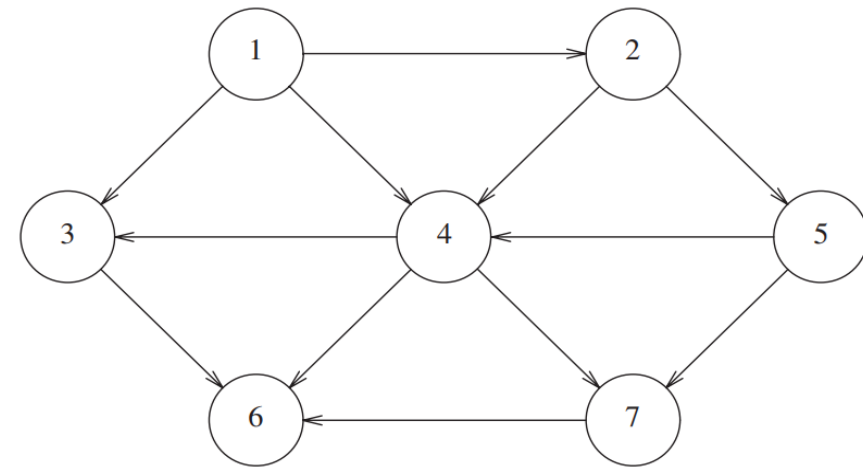
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 - A set of **Edges** (E)
- $V = \{v_1, v_2, \dots, v_n\}$, where total number of vertices is $|V| = n$.



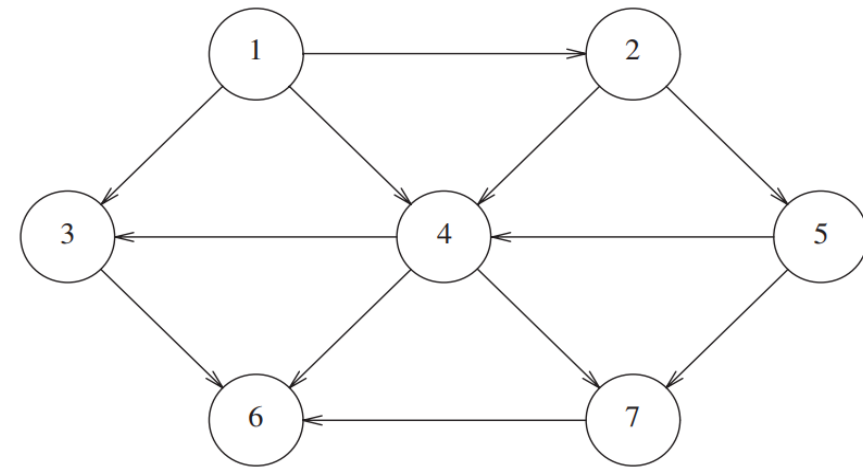
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 - A set of **Vertices** (V)
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- Each edge is a pair (v, w) , where $v, w \in V$.

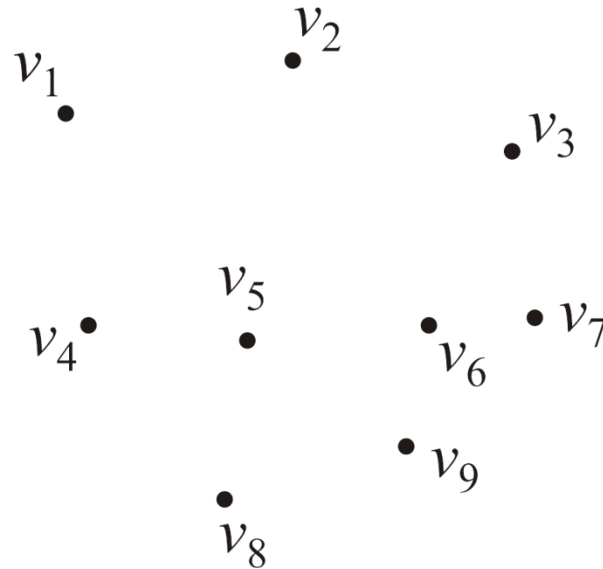


Introduction

- A **Graph** $G = (V, E)$ consists of
 - A set of **Vertices** (V)
 - A set of **Edges** (E)
- $V = \{v_1, v_2, \dots, v_n\}$, where total number of vertices is $|V| = n$.
- Each edge is a pair (v, w) , where $v, w \in V$.
- Graphs can be **directed** or **undirected**.
 - Directed: Edge (v, w) means it starts from v and ends at w .
 - For undirected graph, order is not important.

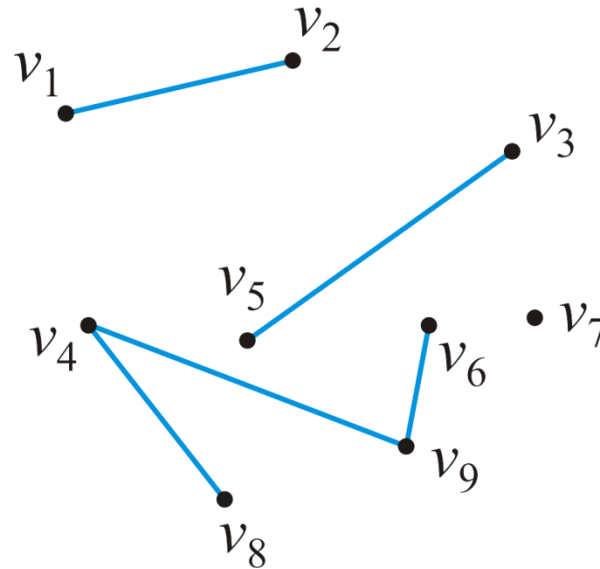


- Consider this collection of vertices $\{v_1, v_2, \dots, v_9\}$



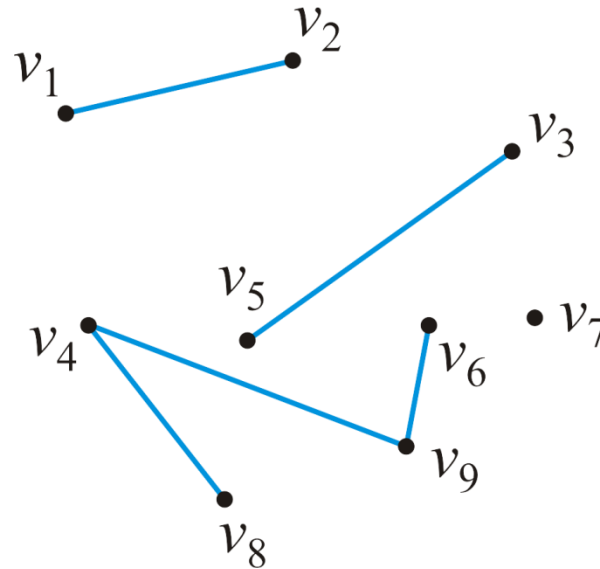
Undirected graphs

- Associated with these vertices are $|E| = 5$ edges
 - $E = \{(v_1, v_2), (v_3, v_5), (v_4, v_8), (v_4, v_9), (v_6, v_9)\}$



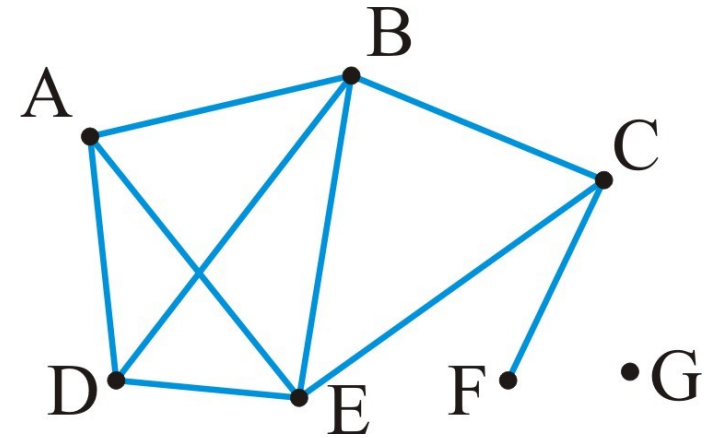
Undirected graphs

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 - $E = \{(v_1, v_2), (v_3, v_5), (v_4, v_8), (v_4, v_9), (v_6, v_9)\}$
- Pair $\{v_j, v_k\}$ indicates: both vertex v_j is adjacent to v_k , and vertex v_k is adjacent to v_j .



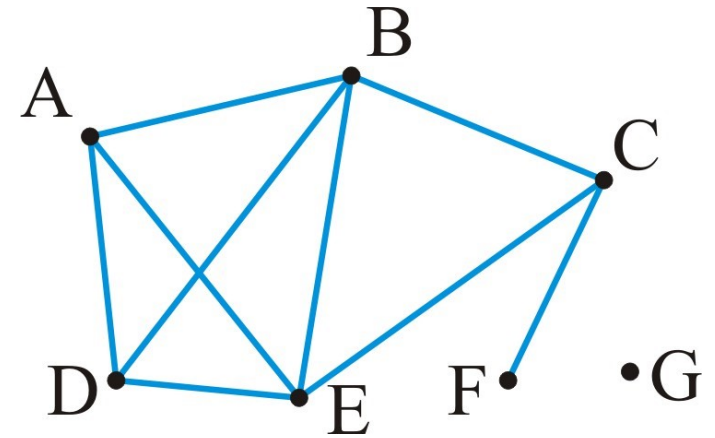
An undirected graph

- Given a graph with
 - $|V| = \{A, B, C, D, E, F, G\}$, 7 vertices
 - $|E| = 9$ edges
 - $E = \{(A, B), (A, D), (A, E), (B, C), (B, D), (B, E), (C, E), (C, F), (D, E)\}$



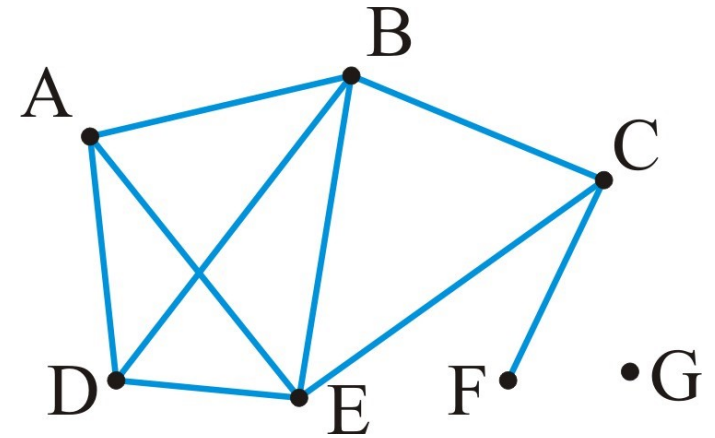
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- The **maximum number of edges** in an undirected graph is
 - ???



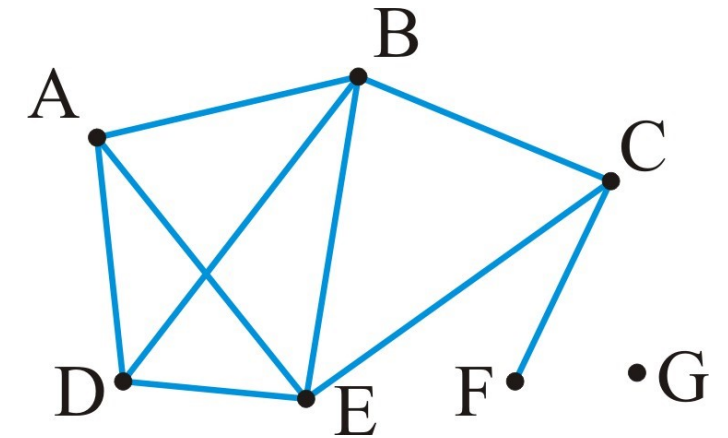
An undirected graph

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 - $|V| = \{A, B, C, D, E, F, G\}$, 7 vertices
 - $|E| = 9$ edges
 - $E = \{(A, B), (A, D), (A, E), (B, C), (B, D), (B, E), (C, E), (C, F), (D, E)\}$
- The **maximum number of edges** in an undirected graph is
 - $E \leq \binom{|V|}{2} = \frac{V(V-1)}{2} = O(V^2)$



Degree

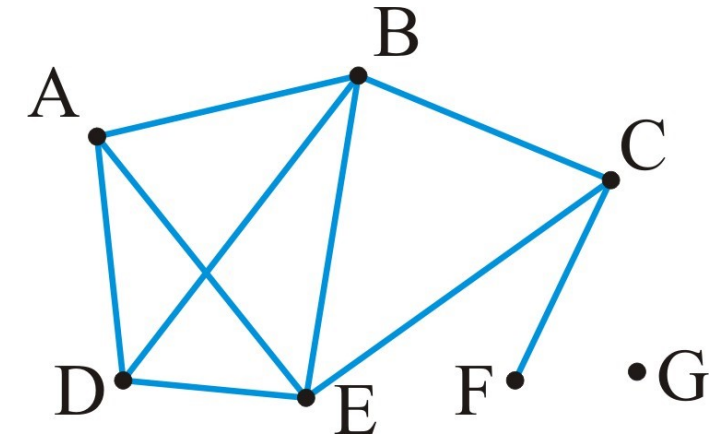
- **Degree** of a vertex = number of adjacent vertices
 - $\text{degree}(A) = \text{degree}(D) = \text{degree}(C) = ???$
 - $\text{degree}(B) = \text{degree}(E) = ???$
 - $\text{degree}(F) = ???$
 - $\text{degree}(G) = ???$



Degree

- **Degree** of a vertex = number of adjacent vertices

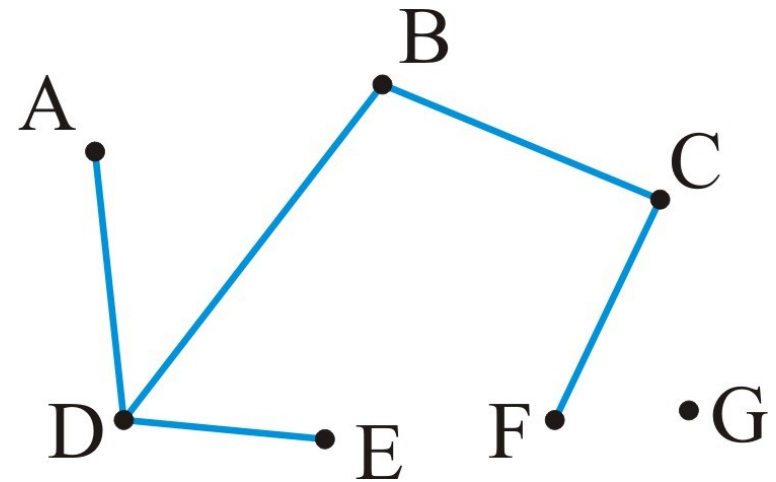
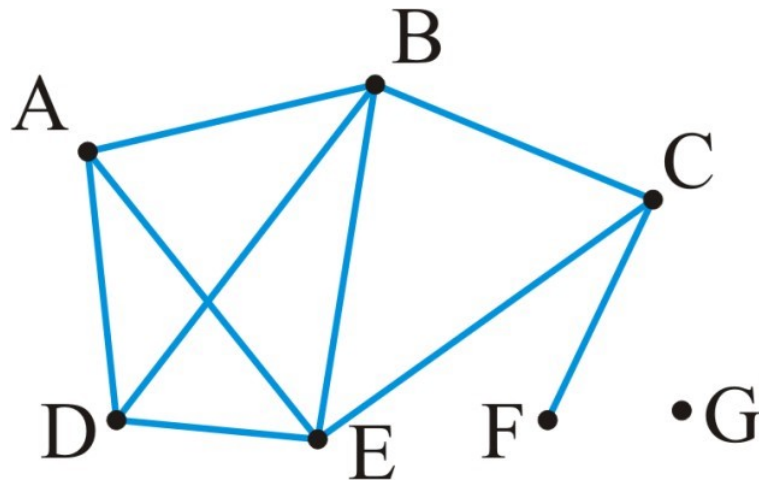
- $\text{degree}(A) = \text{degree}(D) = \text{degree}(C) = 3$
- $\text{degree}(B) = \text{degree}(E) = 4$
- $\text{degree}(F) = 1$
- $\text{degree}(G) = 0$



- The vertices adjacent to a given vertex are its ***neighbors***.

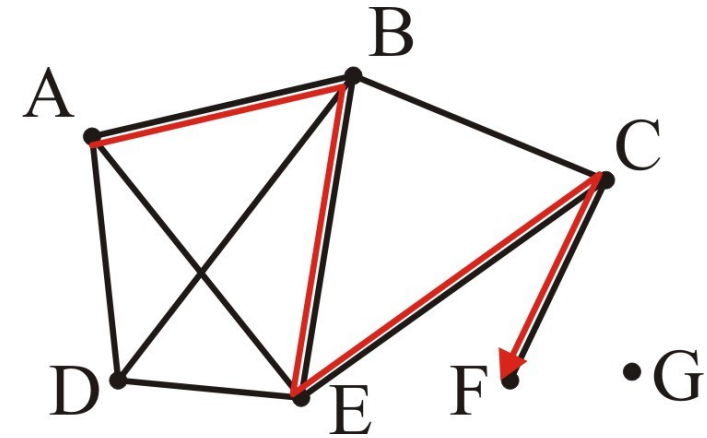
Sub-graphs

- Sub-graph of a graph = a subset of the vertices and a subset of the edges



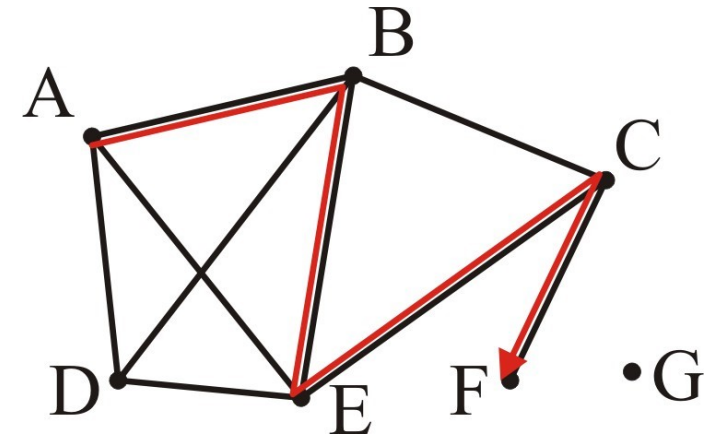
Paths

- Path in an undirected graph: an ordered sequence of vertices $(v_0, v_1, \dots, v_k)$
 - where $\{v_{j-1}, v_j\}$ is an edge for $j = 1, \dots, k$.
 - Termed a path from v_j to v_k
- The length of this path is k



Paths

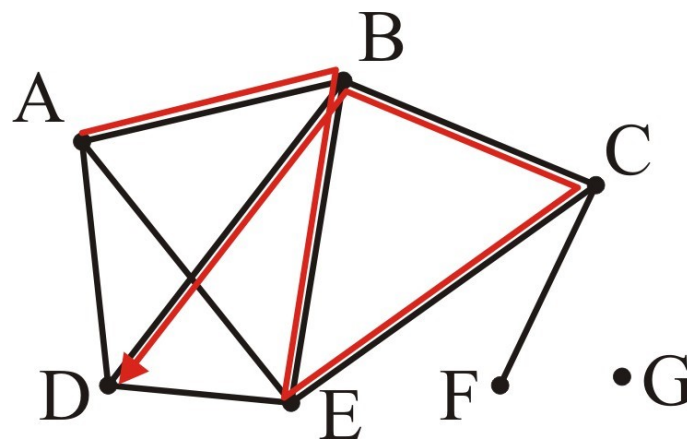
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- **Simple path:** *no repetitions* of vertices
- A simple path of length 4: (A, B, E, C, F)

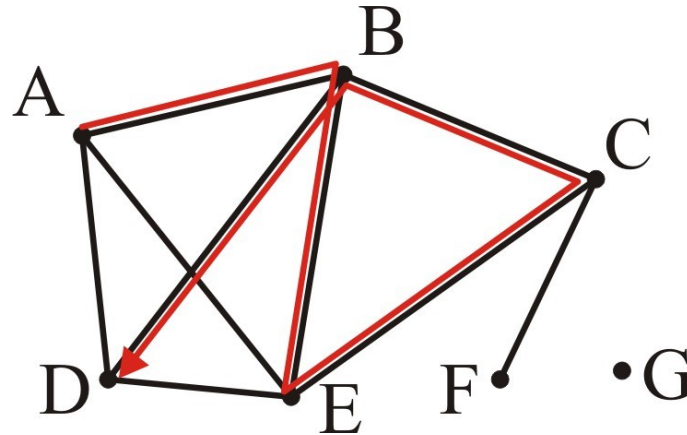
Paths (cont.)

- A path of length 5: (A, B, E, C, B, D)
 - **Is this path simple?**



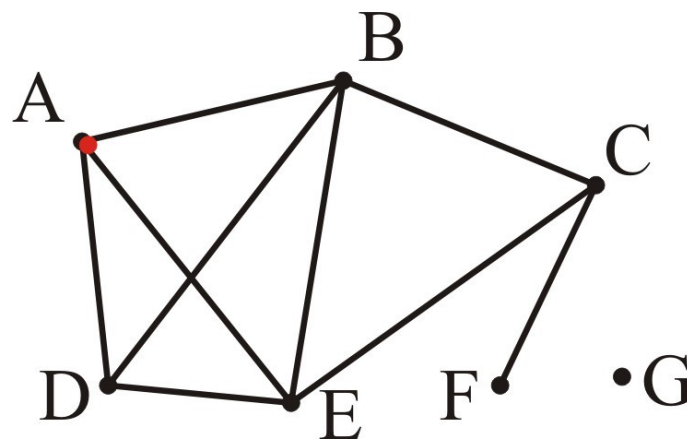
Paths (cont.)

- A path of length 5: (A, B, E, C, B, D)
 - **Is this path simple?**
 - **No !**



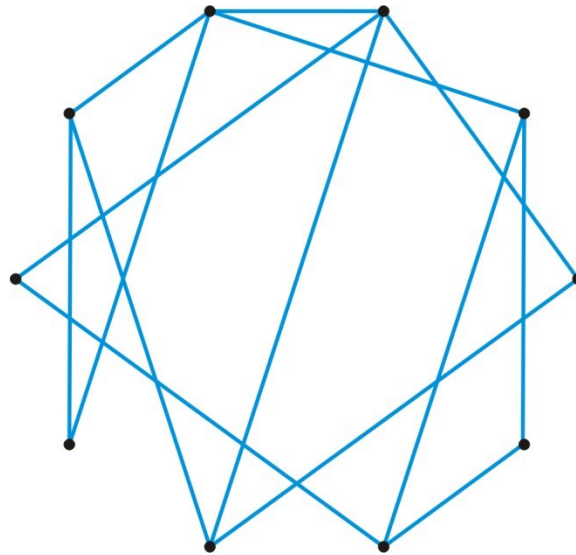
Paths (cont.)

- A trivial path of length 0: (A)

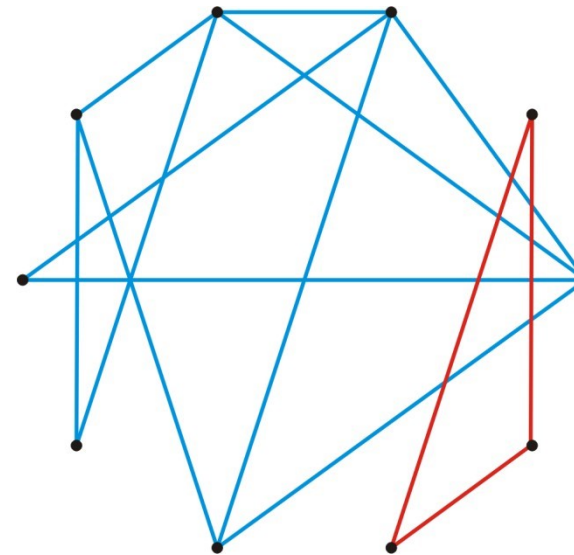


Connectedness

- **Two vertices** v_i, v_j are **connected** if there exists a path from v_i to v_j
- A **graph is connected** if there exists a path between **any two** vertices



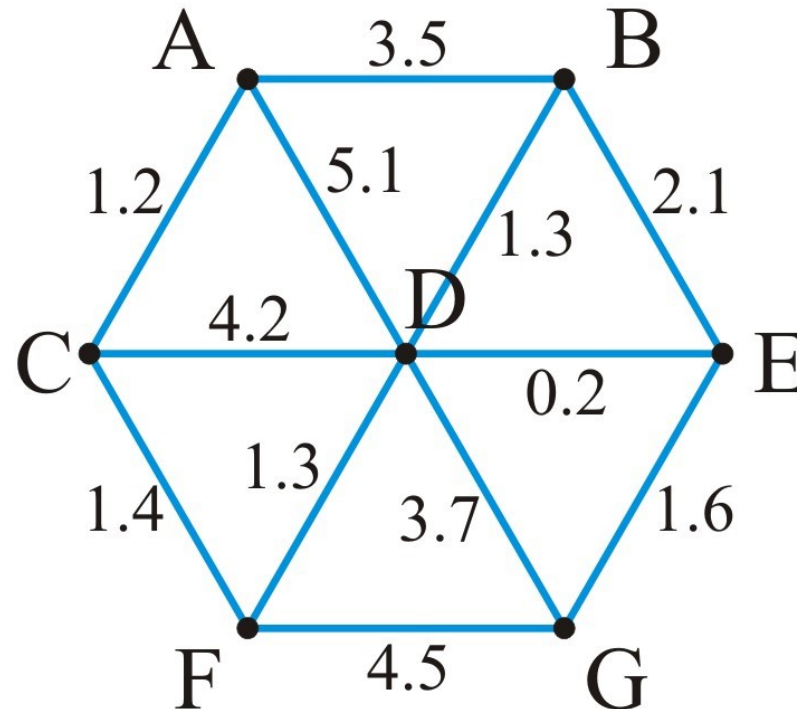
A connected graph



An unconnected graph

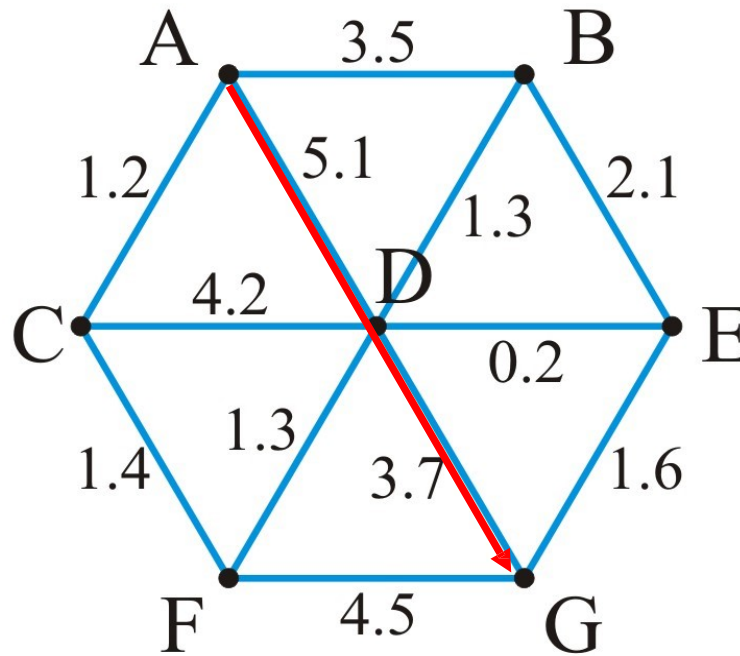
Weighted graphs

- A **weight** may be associated with each edge in a graph
 - Can represent time, distance, energy consumption, cost, etc.
- Such a graph is called a **weighted graph**



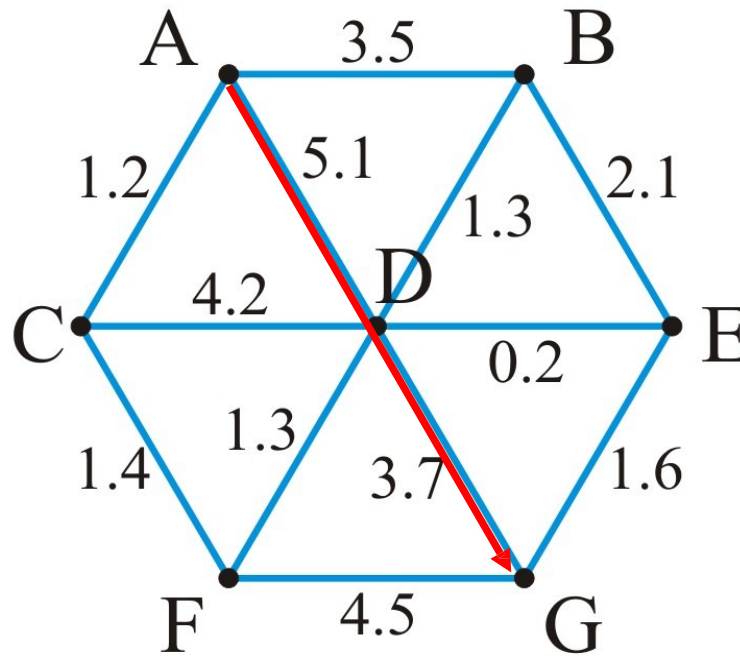
Weighted graphs (cont.)

- **Length of a path** in a weighted graph = sum of all of the edges which make up the path
- The length of the path (A, D, G) is ???



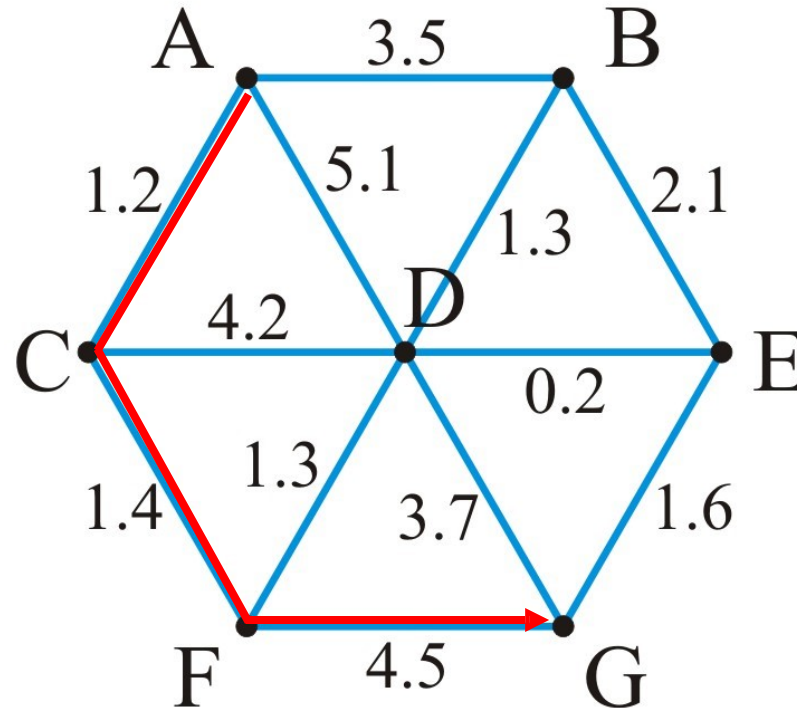
Weighted graphs (cont.)

- **Length of a path** in a weighted graph = sum of all of the edges which make up the path
- The length of the path (A, D, G) is $5.1 + 3.7 = 8.8$



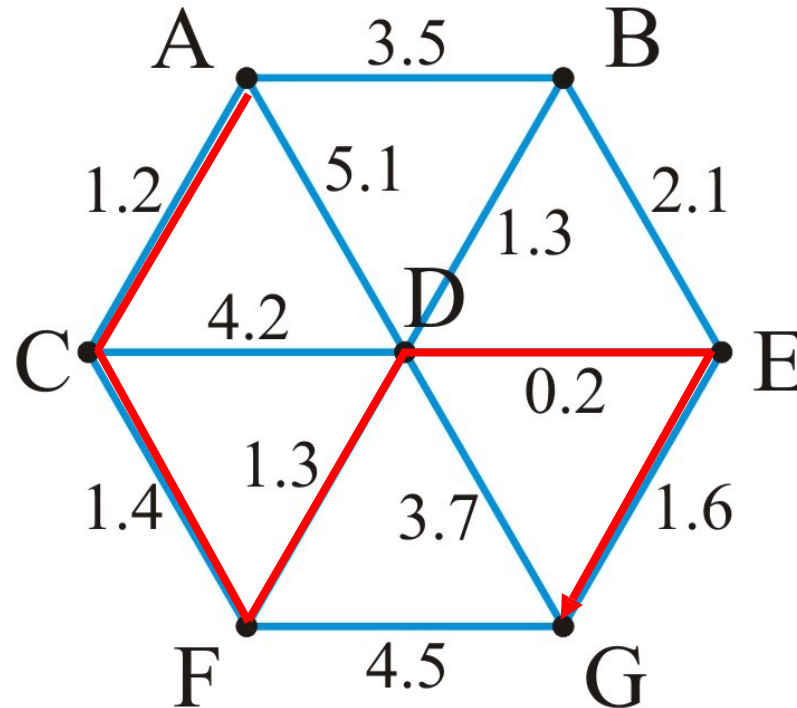
Weighted graphs (cont.)

- Different paths may have different weights
- Another path is (A, C, F, G) with length $1.2 + 1.4 + 4.5 = 7.1$



Weighted graphs (cont.)

- Problem: find the **shortest path** between two vertices
- Here, the shortest path from A to G is (A, C, F, D, E, G) with length 5.7



Directed graphs

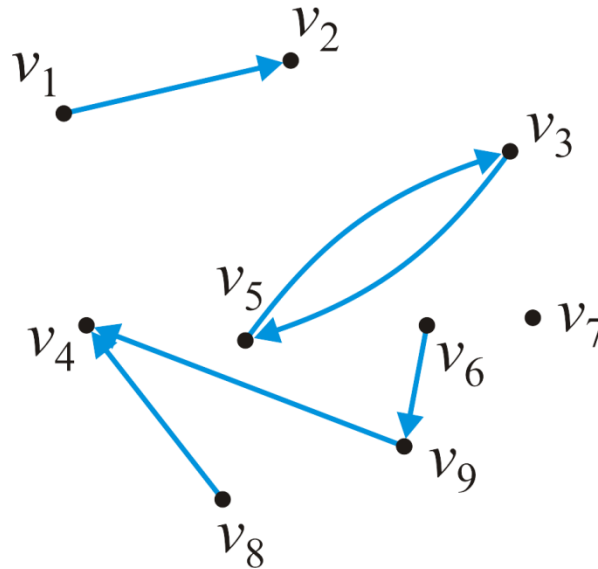
- In a directed graph, the edges are associated with a **direction**
- Edges are **ordered pairs** (v_j, v_k) = connection from v_j to v_k
 - Edge (v_j, v_k) is different from the edge (v_k, v_j)

Directed graphs

- In a directed graph, the edges are associated with a **direction**
- Edges are **ordered pairs** (v_j, v_k) = connection from v_j to v_k
 - Edge (v_j, v_k) is different from the edge (v_k, v_j)
- The maximum number of directed edges in a directed graph is:
 - $E \leq 2 \binom{|V|}{2} = 2 \times \frac{|V|(|V|-1)}{2} = |V|(|V|-1) = O(|V|^2)$

Directed graphs (cont.)

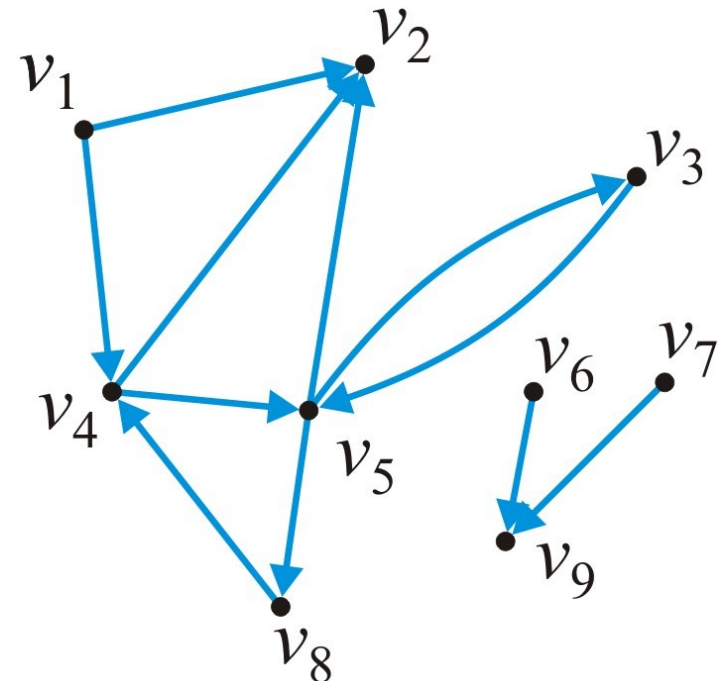
- Given graph of nine vertices $V = \{v_1, v_2, \dots, v_9\}$
- These six pairs (v_j, v_k) are directed edges
- $E = \{(v_1, v_2), (v_3, v_5), (v_5, v_3), (v_6, v_9), (v_8, v_4), (v_9, v_4)\}$



Out-degree & In-degree

- **Out-degree** of a vertex = number of vertices adjacent to the given vertex
- **In-degree** = number of vertices this vertex is adjacent to

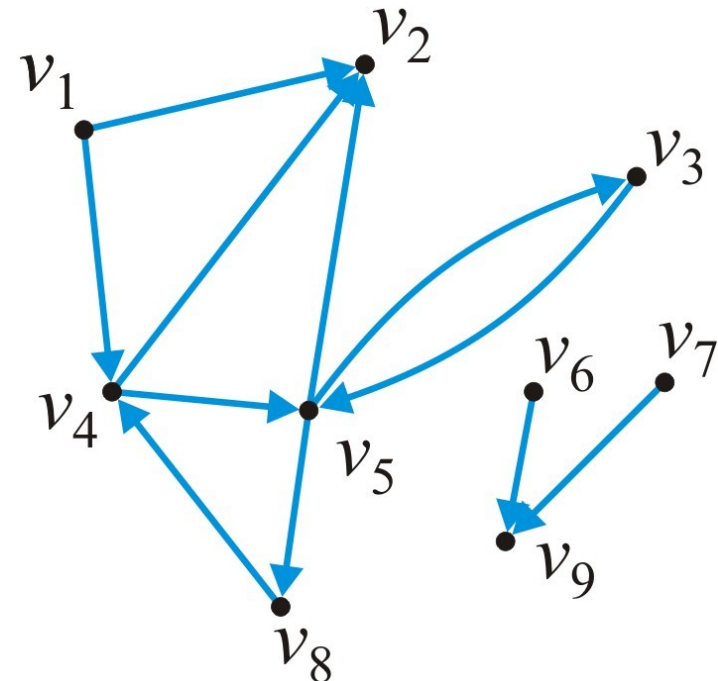
- $inDegree(v_1) = ???$
- $outDegree(v_1) = ???$
- $inDegree(v_5) = ???$
- $outDegree(v_5) = ???$



Out-degree & In-degree

- **Out-degree** of a vertex = number of vertices adjacent to the given vertex
- **In-degree** = number of vertices this vertex is adjacent to

- $inDegree(v_1) = 0$
- $outDegree(v_1) = 2$
- $inDegree(v_5) = 2$
- $outDegree(v_5) = 3$

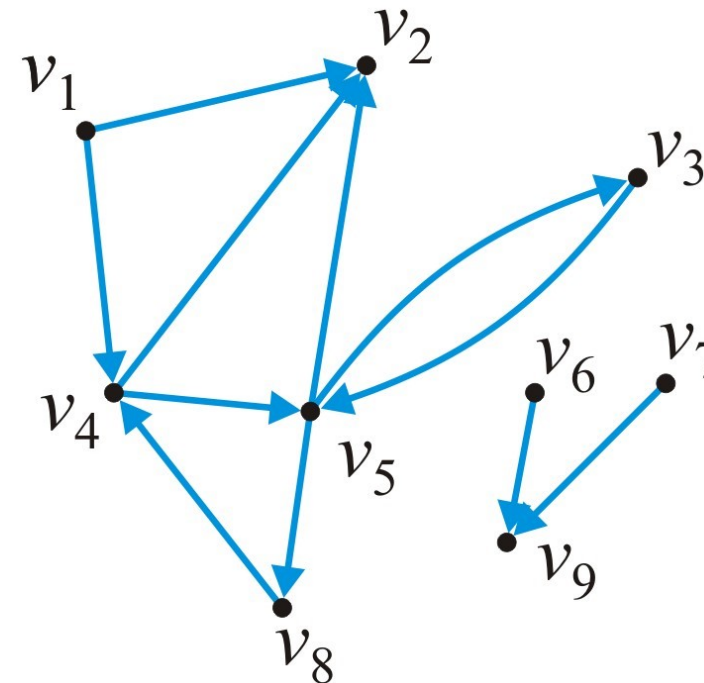


Sources and sinks

- **Sources:** Vertices with an in-degree of zero
- **Sinks:** Vertices with an out-degree of zero

- Sources: ???

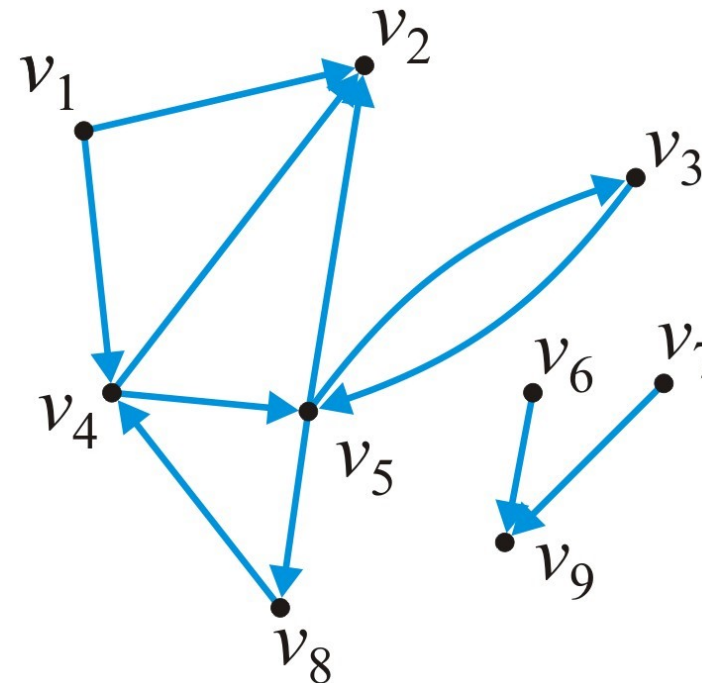
- Sinks: ???



Sources and sinks

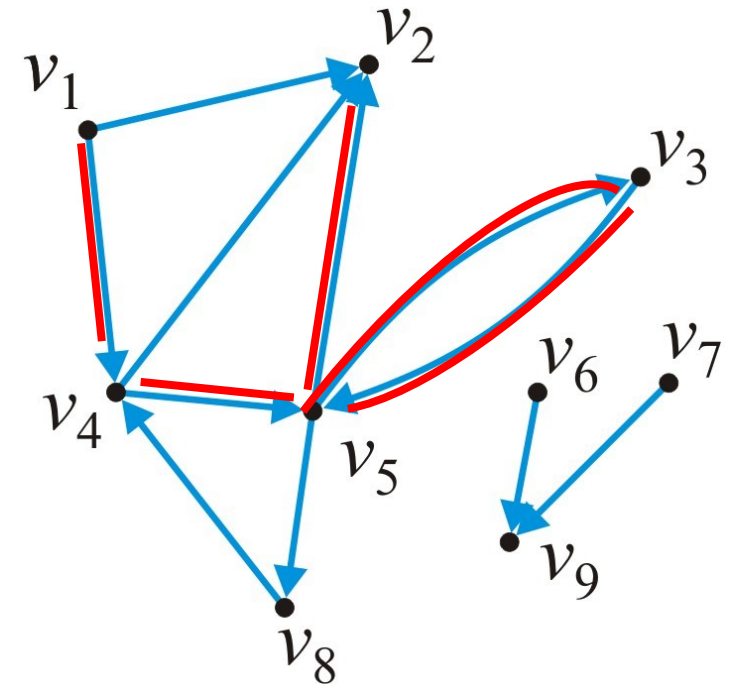
- **Sources:** Vertices with an in-degree of zero
- **Sinks:** Vertices with an out-degree of zero

- Sources: v_1, v_6, v_7
- Sinks: v_2, v_9



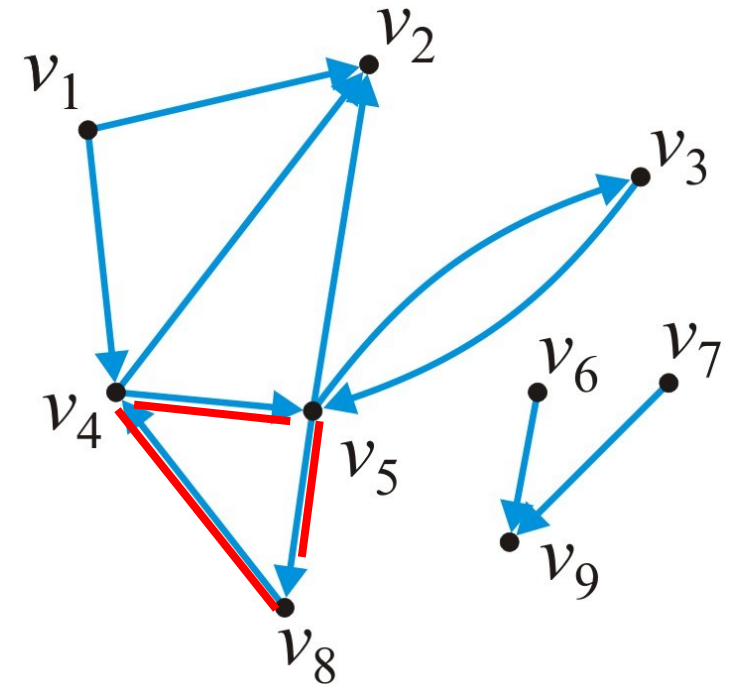
Paths

- Path in a directed graph: ordered sequence of vertices $(v_0, v_1, v_2, \dots, v_k)$
 - where (v_{j-1}, v_j) is an edge for $j = 1, \dots, k$
- Path of length 5: $(v_1, v_4, v_5, v_3, v_5, v_2)$



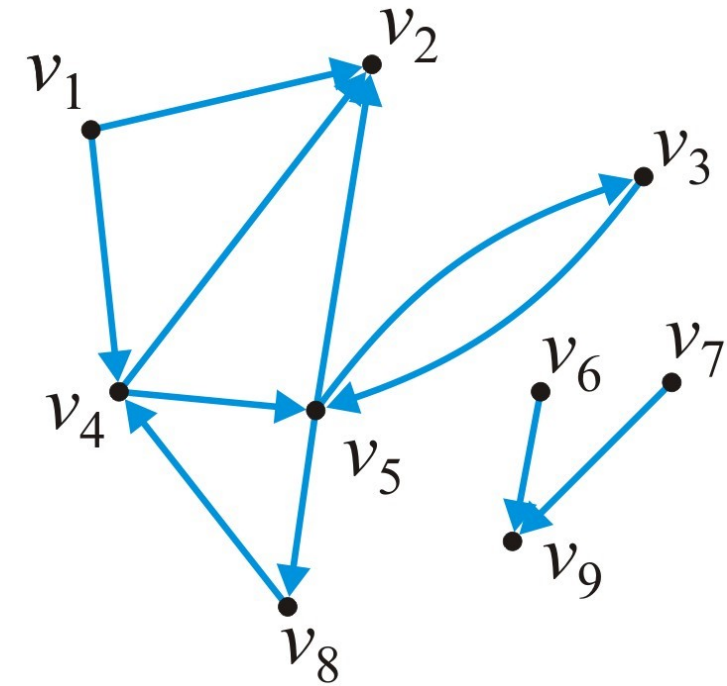
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 - where (v_{j-1}, v_j) is an edge for $j = 1, \dots, k$
- Path of length 5: $(v_1, v_4, v_5, v_3, v_5, v_2)$
- A simple **cycle** of length 3: (v_8, v_4, v_5, v_8)



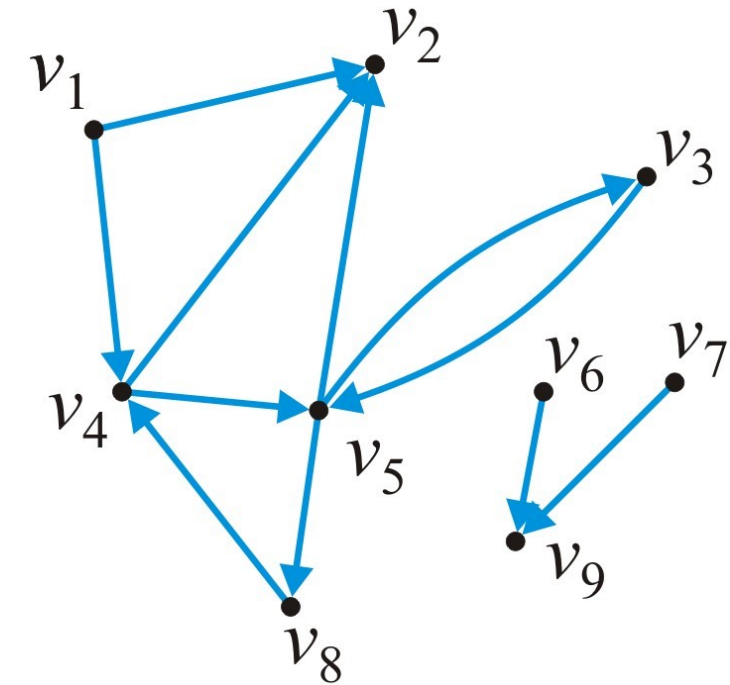
Connectedness

- Two vertices v_j, v_k are connected if there exists a path from v_j to v_k



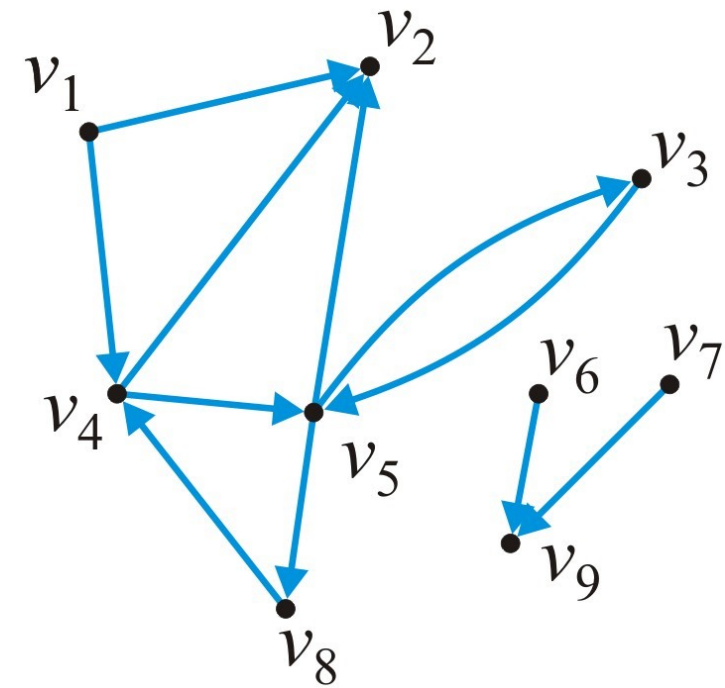
Connectedness

- Two vertices v_j, v_k are connected if there exists a path from v_j to v_k
- A graph is **strongly connected** if there exists a **directed path** between any two vertices



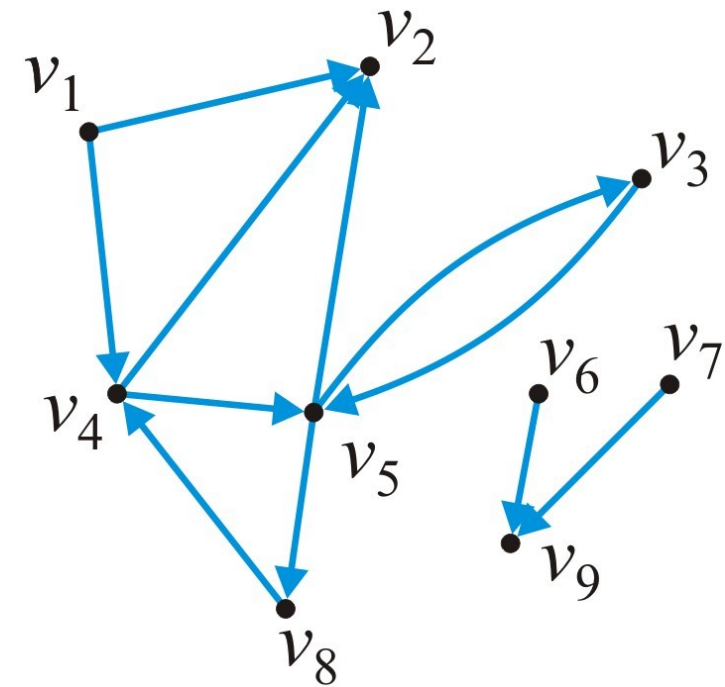
Connectedness

- A directed graph is **strongly connected** if there is a **directed path** from any vertex to any other vertex.
- If we suppress the direction of the edges and the resulting undirected graph is connected, we call the directed graph **weakly connected**.



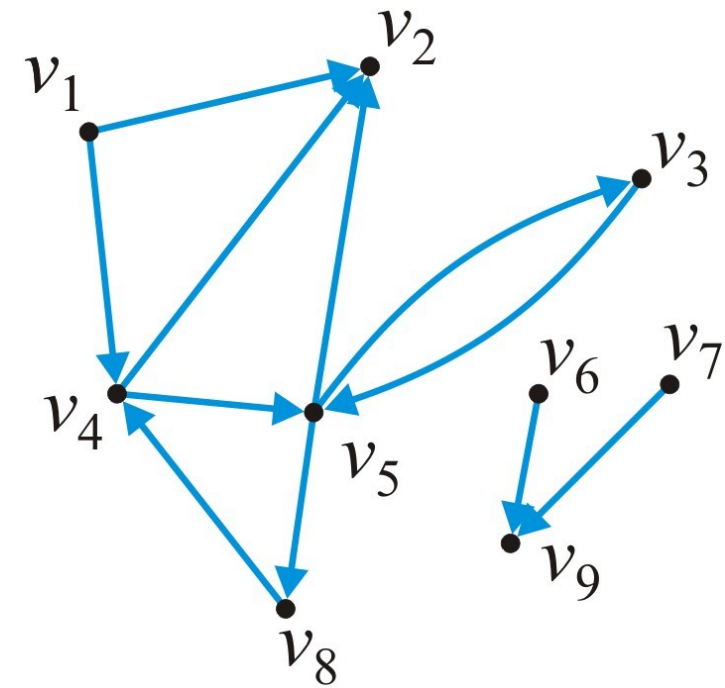
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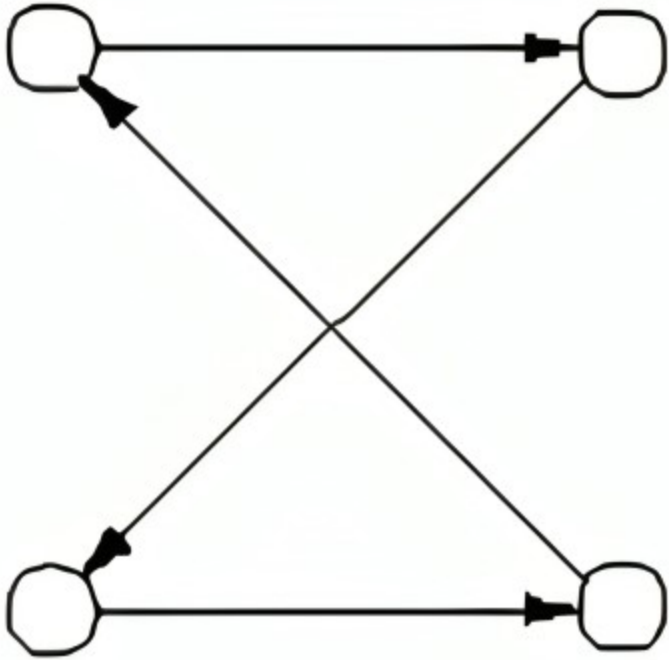
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- Sub-graph $\{v_3, v_4, v_5, v_8\}$ is ??? (strongly/weakly) connected
- Sub-graph $\{v_1, v_2, v_3, v_4, v_5, v_8\}$ is ??? (strongly/weakly) connected.



Connectedness

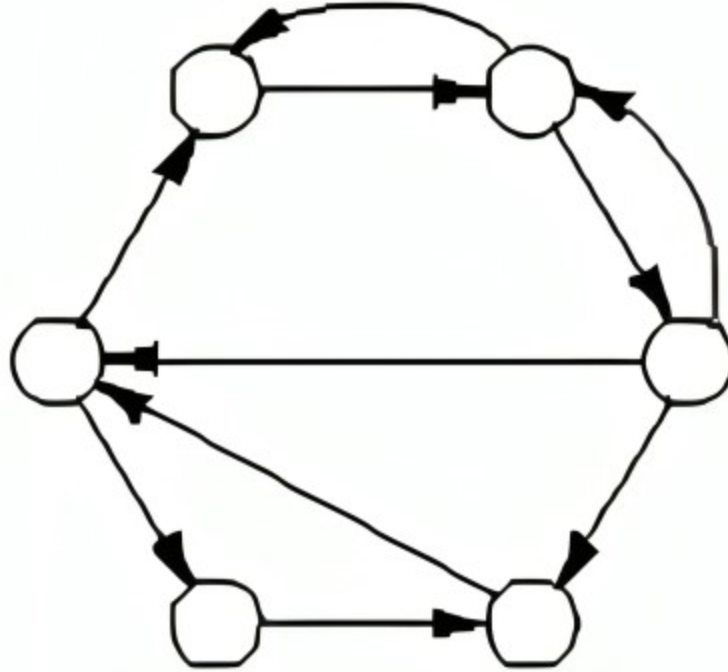
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- If we suppress the direction of the edges and the resulting undirected graph is connected, we call the directed graph **weakly connected**.
- Sub-graph $\{v_3, v_4, v_5, v_8\}$ is **strongly** connected
- Sub-graph $\{v_1, v_2, v_3, v_4, v_5, v_8\}$ is **weakly** connected.





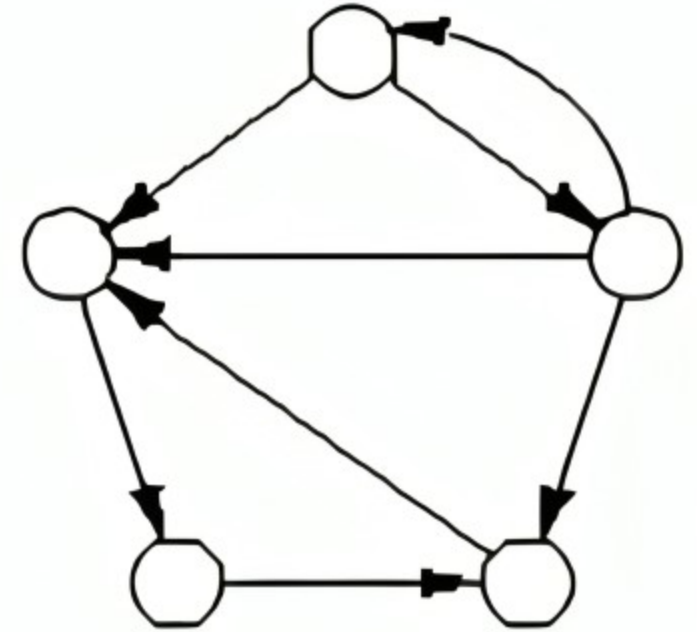
Directed cycle

(a)



Strongly connected

(b)

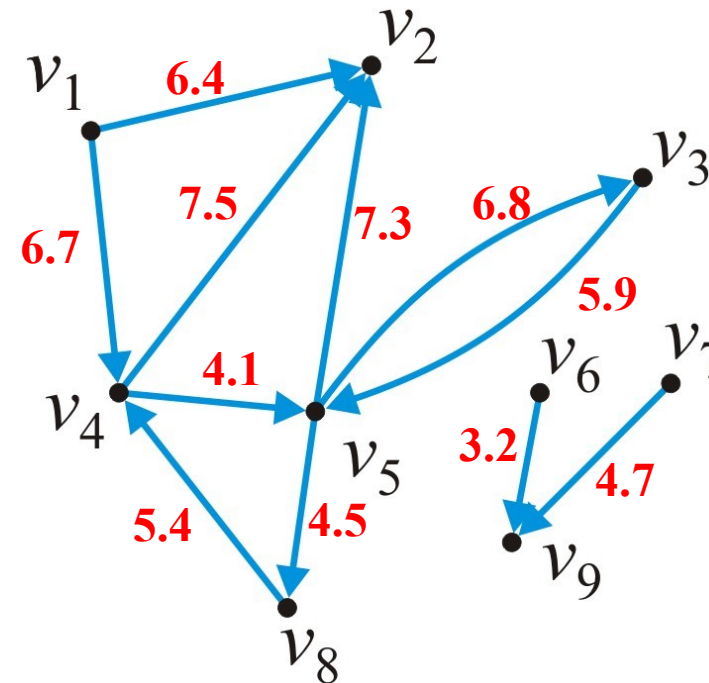


Weakly connected

(c)

Weighted directed graphs

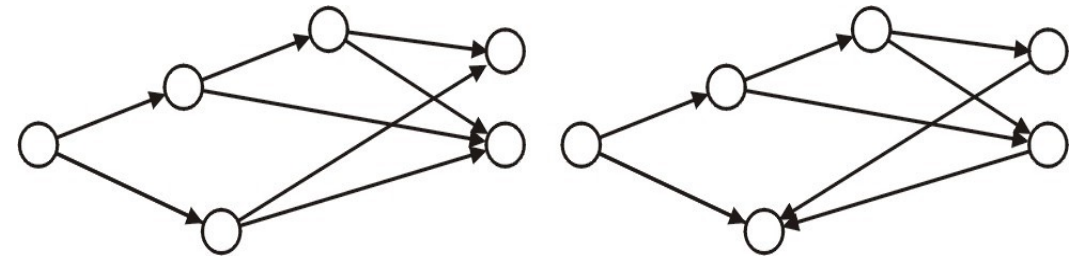
- In a weighted directed graph, each edge is associated with a value
- Unlike weighted undirected graphs:
 - If both (v_j, v_k) and (v_k, v_j) are edges, it is **not required that they have the same weight**



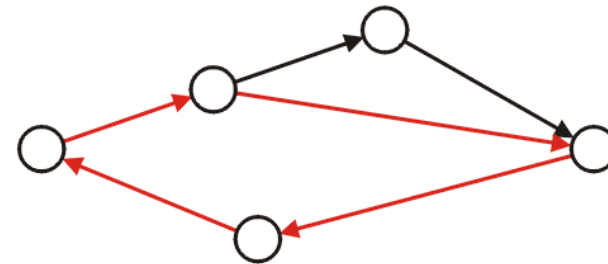
Directed acyclic graphs

- **Directed acyclic graph** is a directed graph which has no cycles (**DAGs**)

- These two are DAGs:

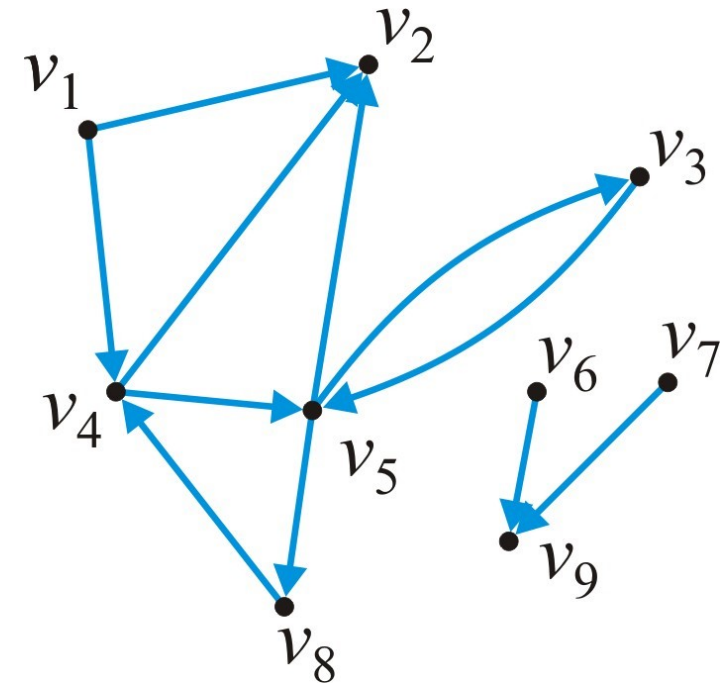


- This directed graph is not acyclic:



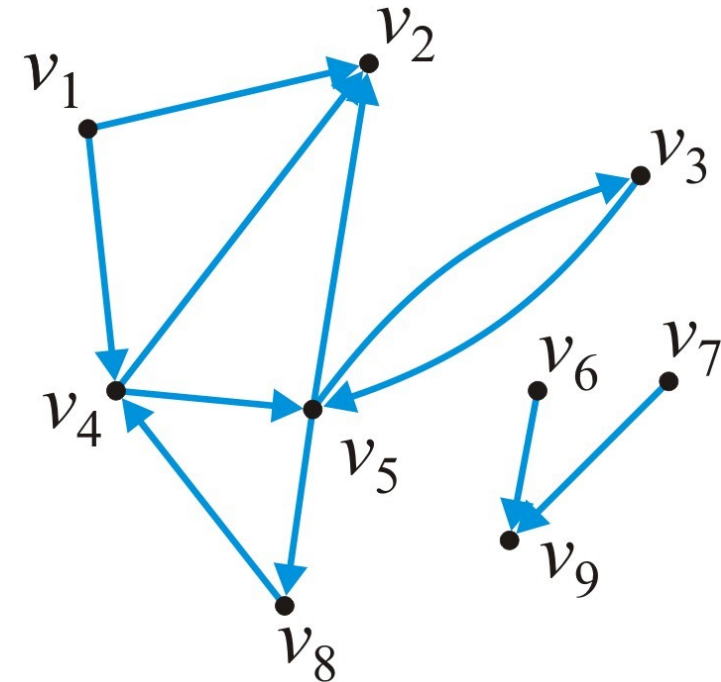
Representations

- How do we store the adjacency relations?
 - Binary-relation list
 - Adjacency matrix
 - Adjacency list



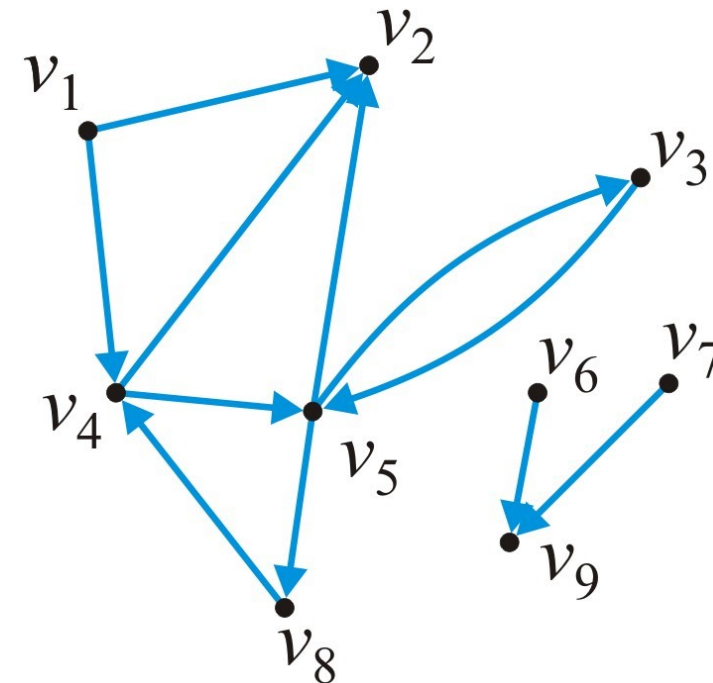
Binary-relation list

- The **most inefficient** is a relation list:
 - A container storing the edges
 - $\{(1, 2), (1, 4), (3, 5), (4, 2), (4, 5), (5, 2), (5, 3), (5, 8), (6, 9), (7, 9), (8, 4)\}$
- Requires $O(???)$ memory
- Determine if v_j is adjacent to v_k : $O(???)$
- Find all neighbors of v_j : $O(???)$



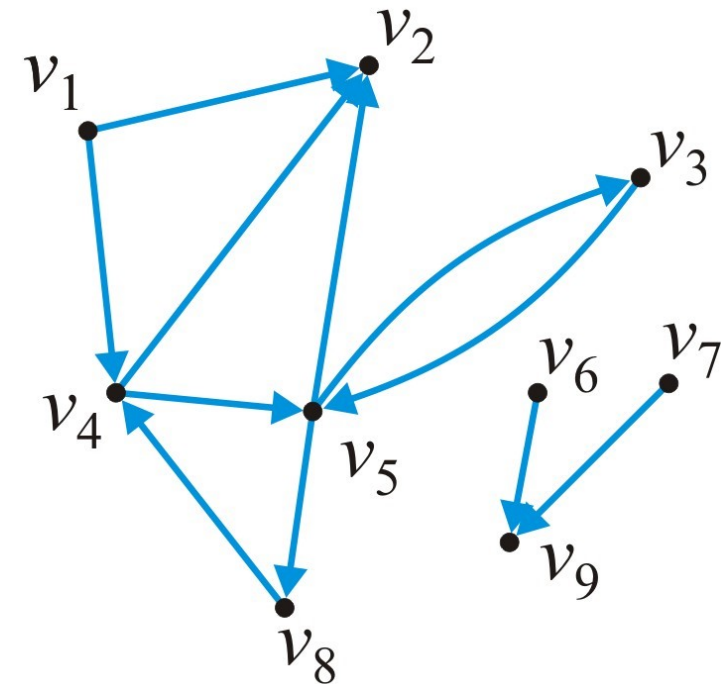
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- Requires $O(|E|)$ memory
- Determine if v_j is adjacent to v_k : $O(???)$
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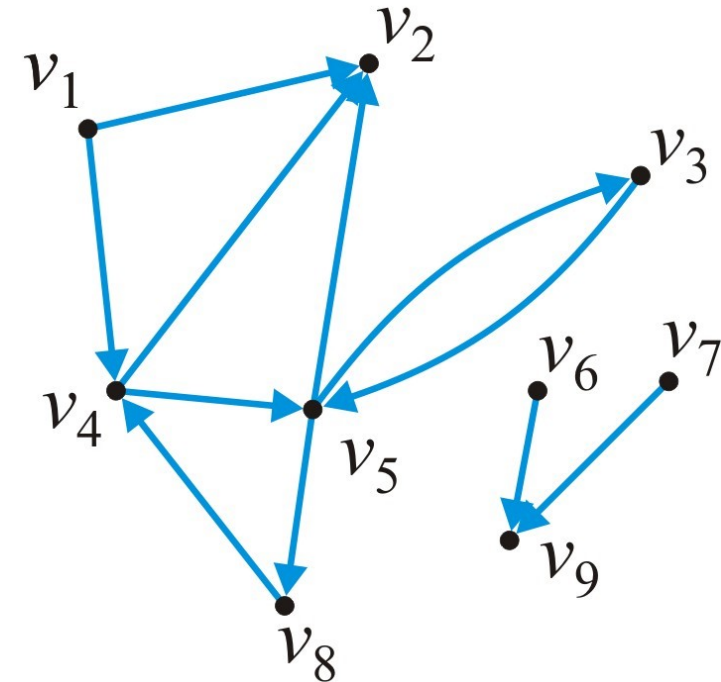
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- Requires $O(|E|)$ memory
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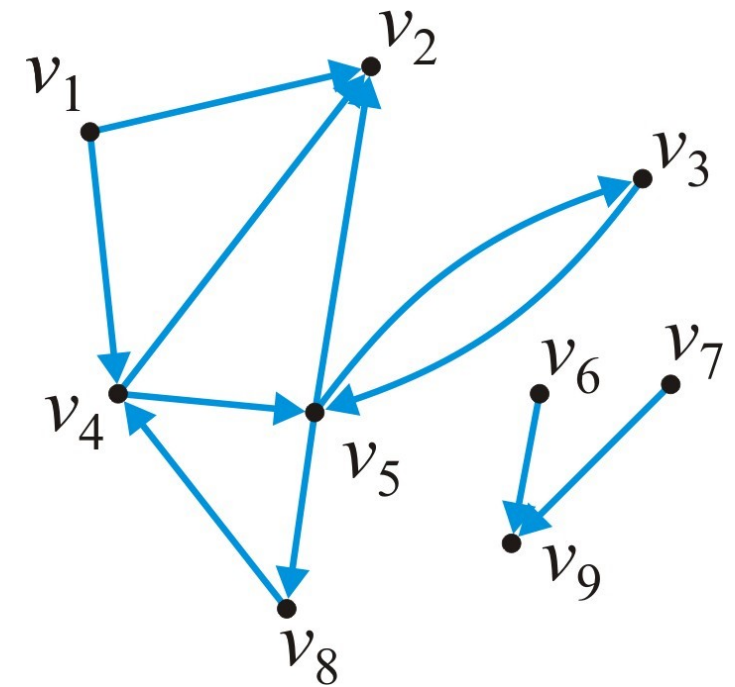
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- Requires $O(|E|)$ memory
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- Find all neighbors of v_j : $O(|E|)$



Adjacency matrix

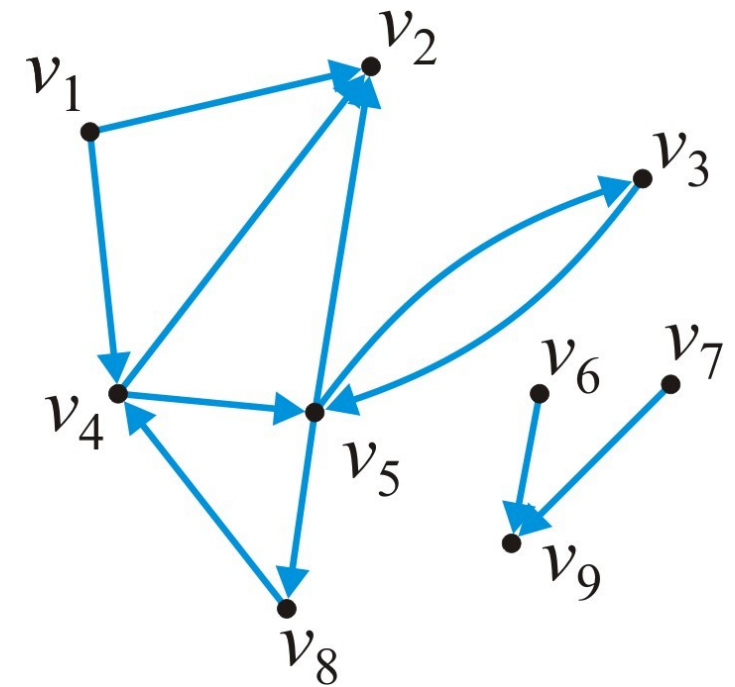
- Requiring **more memory** but also **faster**, an adjacency matrix
- The matrix entry (j, k) is set to true if there is an edge (v_j, v_k)
- Requires $O(???)$ memory
- Determine if v_j is adjacent to v_k : $O(???)$
- Find all neighbors of v_j : $O(???)$



	1	2	3	4	5	6	7	8	9
1		T		T					
2									
3					T				
4		T			T				
5		T	T					T	
6									T
7									T
8				T					
9									

Adjacency matrix

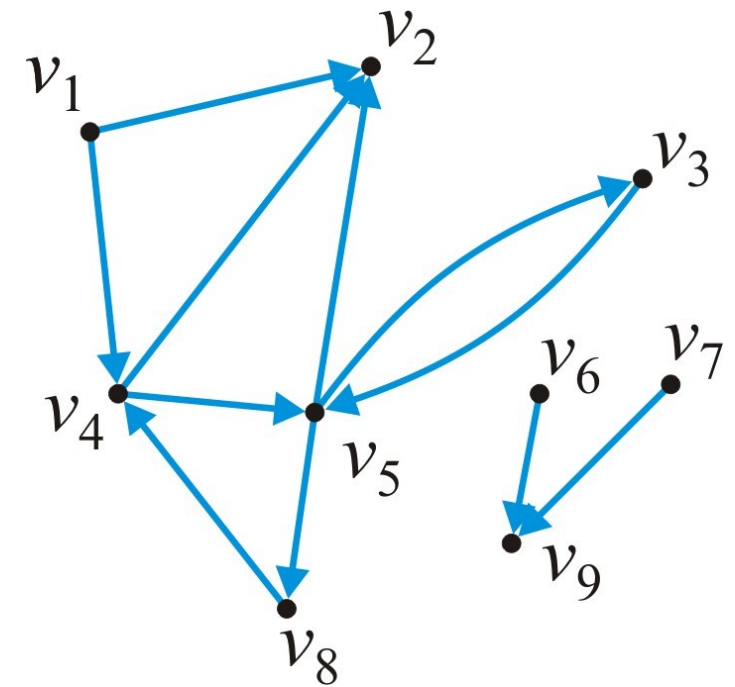
- Requiring **more memory** but also **faster**, an adjacency matrix
- The matrix entry (j, k) is set to true if there is an edge (v_j, v_k)
- Requires $O(|V|^2)$ memory
- Determine if v_j is adjacent to v_k : $O(???)$
- Find all neighbors of v_j : $O(???)$



	1	2	3	4	5	6	7	8	9
1		T		T					
2									
3					T				
4		T			T				
5		T	T					T	
6									T
7									T
8				T					
9									

Adjacency matrix

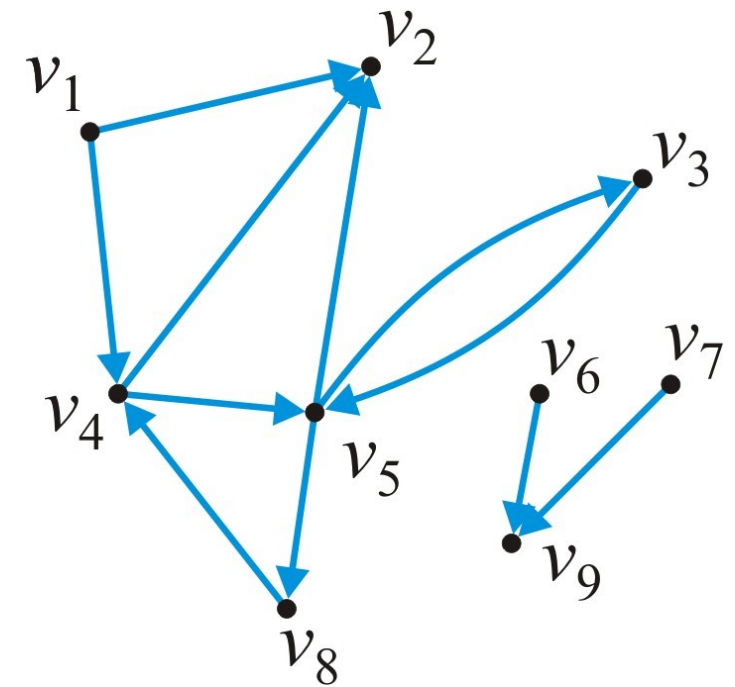
- Requiring **more memory** but also **faster**, an adjacency matrix
- The matrix entry (j, k) is set to true if there is an edge (v_j, v_k)
- Requires $O(|V|^2)$ memory
- Determine if v_j is adjacent to v_k : $O(1)$
- Find all neighbors of v_j : $O(???)$



	1	2	3	4	5	6	7	8	9
1		T		T					
2									
3					T				
4		T			T				
5		T	T					T	
6									T
7									T
8				T					
9									

Adjacency matrix

- Requiring **more memory** but also **faster**, an adjacency matrix
- The matrix entry (j, k) is set to true if there is an edge (v_j, v_k)
- Requires $O(|V|^2)$ memory
- Determine if v_j is adjacent to v_k : $O(1)$
- Find all neighbors of v_j : $O(|V|)$



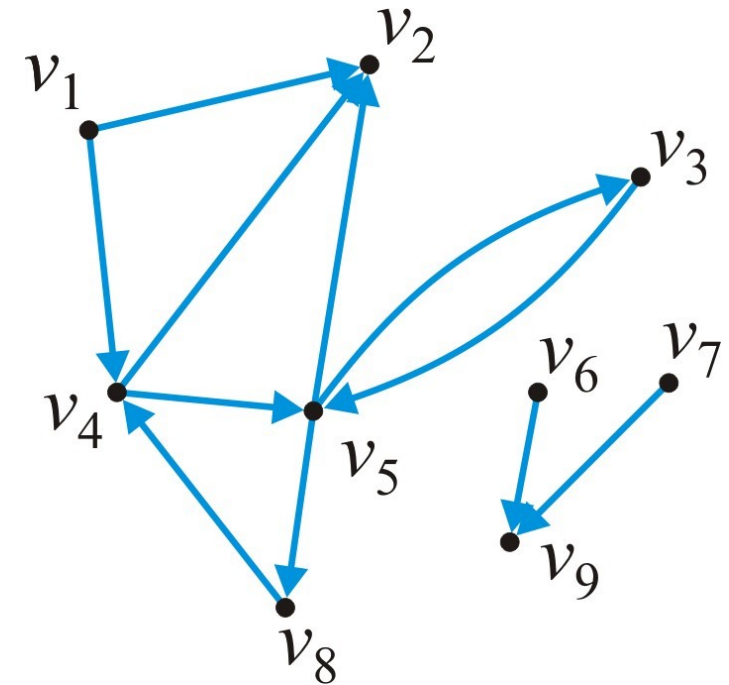
	1	2	3	4	5	6	7	8	9
1		T		T					
2									
3					T				
4		T			T				
5		T	T					T	
6									T
7									T
8				T					
9									

Adjacency list

- **Most efficient** for algorithms is an adjacency list
- Each vertex is associated with a list of its neighbors

1 • $\rightarrow$ 2 $\rightarrow$ 4
2 •
3 • $\rightarrow$ 5
4 • $\rightarrow$ 2 $\rightarrow$ 5
5 • $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 8
6 • $\rightarrow$ 9
7 • $\rightarrow$ 9
8 • $\rightarrow$ 4
9 •

- Requires $O(???)$ memory
- On average:
 - Determining if v_j is adjacent to v_k is $O(???)$
 - Find all neighbors of v_j is $O(???)$

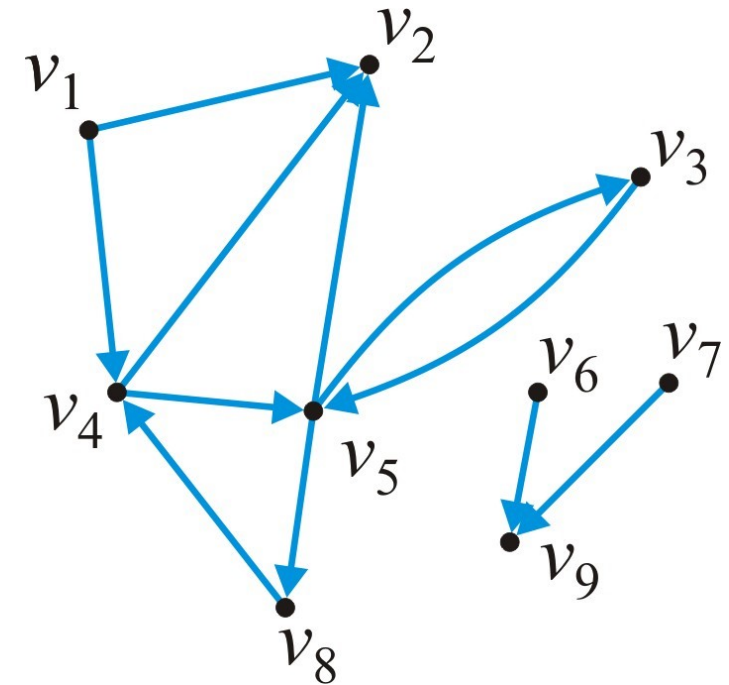


Adjacency list

- **Most efficient** for algorithms is an adjacency list
- Each vertex is associated with a list of its neighbors

1 • $\rightarrow$ 2 $\rightarrow$ 4
2 •
3 • $\rightarrow$ 5
4 • $\rightarrow$ 2 $\rightarrow$ 5
5 • $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 8
6 • $\rightarrow$ 9
7 • $\rightarrow$ 9
8 • $\rightarrow$ 4
9 •

- Requires $O(|V| + |E|)$ memory
- On average:
 - Determining if v_j is adjacent to v_k is $O(???)$
 - Find all neighbors of v_j is $O(???)$

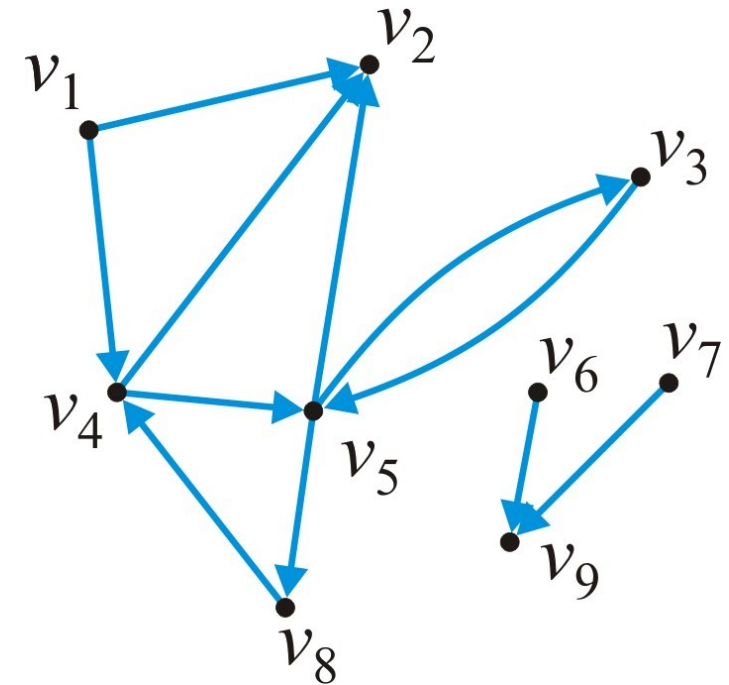


Adjacency list

- **Most efficient** for algorithms is an adjacency list
- Each vertex is associated with a list of its neighbors

1 • $\rightarrow$ 2 $\rightarrow$ 4
2 •
3 • $\rightarrow$ 5
4 • $\rightarrow$ 2 $\rightarrow$ 5
5 • $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 8
6 • $\rightarrow$ 9
7 • $\rightarrow$ 9
8 • $\rightarrow$ 4
9 •

- Requires $O(|V| + |E|)$ memory
- On average:
 - Determining if v_j is adjacent to v_k is $O(\text{avg length of list})$
 - Find all neighbors of v_j is $O(\text{avg length of list})$

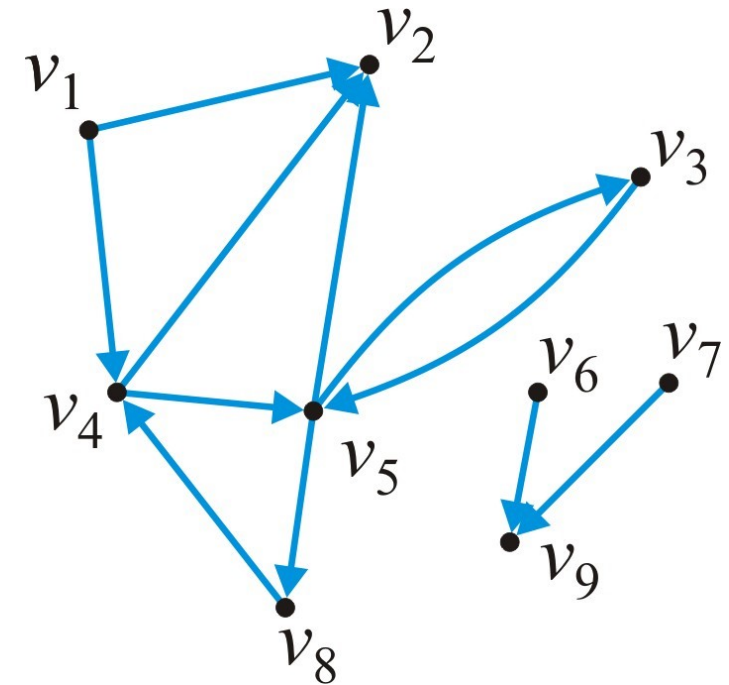


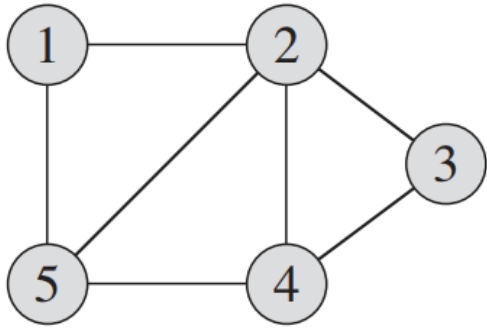
Adjacency list

- **Most efficient** for algorithms is an adjacency list
- Each vertex is associated with a list of its neighbors

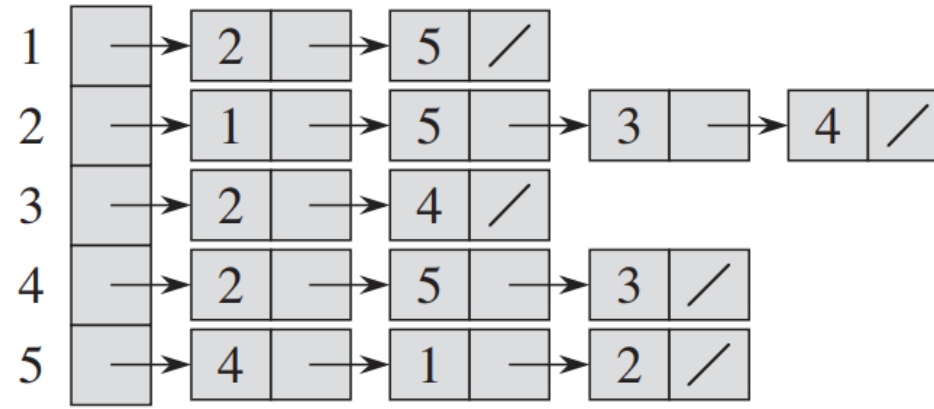
1 • $\rightarrow$ 2 $\rightarrow$ 4
2 •
3 • $\rightarrow$ 5
4 • $\rightarrow$ 2 $\rightarrow$ 5
5 • $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 8
6 • $\rightarrow$ 9
7 • $\rightarrow$ 9
8 • $\rightarrow$ 4
9 •

- Requires $O(|V| + |E|)$ memory
- On average:
 - Determining if v_j is adjacent to v_k is $O(|E|/|V|)$
 - Find all neighbors of v_j is $O(|E|/|V|)$





(a)

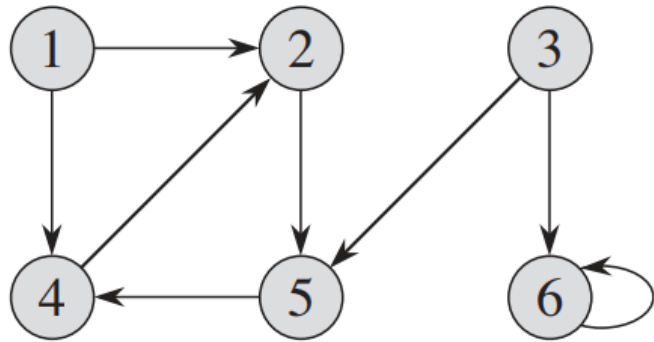


(b)

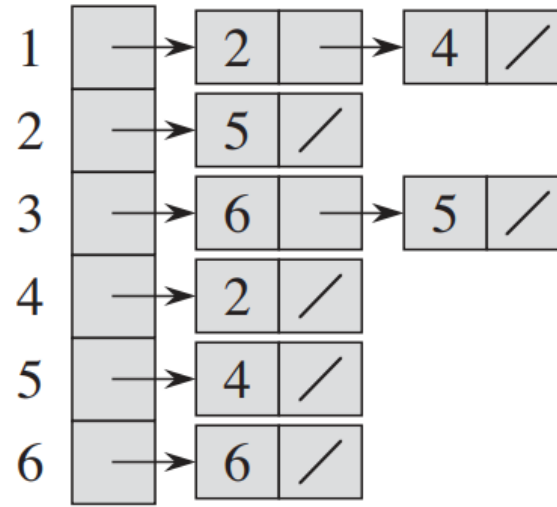
	1	2	3	4	5
1	0	1	0	0	1
2	1	0	1	1	1
3	0	1	0	1	0
4	0	1	1	0	1
5	1	1	0	1	0

(c)

Figure 22.1 Two representations of an undirected graph. (a) An undirected graph G with 5 vertices and 7 edges. (b) An adjacency-list representation of G . (c) The adjacency-matrix representation of G .



(a)



(b)

	1	2	3	4	5	6
1	0	1	0	1	0	0
2	0	0	0	0	1	0
3	0	0	0	0	1	1
4	0	1	0	0	0	0
5	0	0	0	1	0	0
6	0	0	0	0	0	1

(c)

Figure 22.2 Two representations of a directed graph. **(a)** A directed graph G with 6 vertices and 8 edges. **(b)** An adjacency-list representation of G . **(c)** The adjacency-matrix representation of G .

Adjacency Matrix vs Adjacency List

- Two common graph operations:
 1. Determine **whether there is an edge** from v_i to v_j .
 2. Find **all vertices adjacent** to a given vertex v_i .

Adjacency matrix	Adjacency list

Adjacency Matrix vs Adjacency List

- Two common graph operations:
 1. Determine **whether there is an edge** from v_i to v_j .
 2. Find **all vertices adjacent** to a given vertex v_i .

Adjacency matrix	Adjacency list
Supports operation 1 more efficiently	

Adjacency Matrix vs Adjacency List

- Two common graph operations:
 1. Determine **whether there is an edge** from v_i to v_j .
 2. Find **all vertices adjacent** to a given vertex v_i .

Adjacency matrix	Adjacency list
Supports operation 1 more efficiently	Supports operation 2 more efficiently

Adjacency Matrix vs Adjacency List

- Two common graph operations:
 1. Determine **whether there is an edge** from v_i to v_j .
 2. Find **all vertices adjacent** to a given vertex v_i .

Adjacency matrix	Adjacency list
Supports operation 1 more efficiently	Supports operation 2 more efficiently
Space requirement: $O(???)$	Space requirement: $O(???)$

Adjacency Matrix vs Adjacency List

- Two common graph operations:
 1. Determine **whether there is an edge** from v_i to v_j .
 2. Find **all vertices adjacent** to a given vertex v_i .

Adjacency matrix	Adjacency list
Supports operation 1 more efficiently	Supports operation 2 more efficiently
Space requirement: $O(V ^2)$	Space requirement: $O(V + E)$, linear in the size of the graph

Adjacency Matrix vs Adjacency List

- Two common graph operations:
 1. Determine **whether there is an edge** from v_i to v_j .
 2. Find **all vertices adjacent** to a given vertex v_i .

Adjacency matrix	Adjacency list
Supports operation 1 more efficiently	Supports operation 2 more efficiently
Space requirement: $O(V ^2)$	Space requirement: $O(V + E)$, linear in the size of the graph
Better if the graph is dense/sparse ?	Better if the graph is dense/sparse ?

Adjacency Matrix vs Adjacency List

- Two common graph operations:
 1. Determine **whether there is an edge** from v_i to v_j .
 2. Find **all vertices adjacent** to a given vertex v_i .

Adjacency matrix	Adjacency list
Supports operation 1 more efficiently	Supports operation 2 more efficiently
Space requirement: $O(V ^2)$	Space requirement: $O(V + E)$, linear in the size of the graph
Better if the graph is dense (too many edges)	Better if the graph is sparse (few edges)

Remarks

- Adjacency-list: **compact** way to represent **sparse graphs** ($E \ll |V^2|$).
 - (Most graph algo presented in the CLRS book assume Adj-list in the backend)
- Adjacency matrix: may be prefer for **dense graphs** ($E \approx |V^2|$).
 - Can quickly tell if two vertices are connected via an Edge $O(1)$.
 - Adj-matrix A of an undirected graph is its own transpose: $A = A^T$.
 - In some applications, it stores only the entries on and above the diagonal of the matrix, thereby cutting the memory requirement almost in half.
- How can we store weighted graphs in Adj-list & Adj-matrix?

- Reference text:
 - Chapter 22, CLRS
 - Chapter 9, Data Structures & Algorithm Analysis in C++, 4th Edition, by Mark Weiss